# Topic 3 — Food Chain (खाद्य श्रृंखला) & Energy Flow

**Covers syllabus (click to expand)**

Food Chain | Food Web (खाद्य जाल) | Trophic Levels | Energy Flow in Ecosystem (पारिस्थितिकी तंत्र) | Ecological Pyramid (पिरामिड) (Ecological Pyramids)

**Sources baked in:** NCERT Class 12 Biology Ch 14, Odum, Elton (1927), Lindeman (1942), UPPCS/UPSC PYQs 2018–2025 **Weight:** ★★★ Very High — direct Prelims; links to pollution (biomagnification) **Last verified:** August 2026

## Current Affairs (this topic)

- Energy-flow and trophic-level traps recycle yearly — 10% rule and pyramid types stay core.
- Bioaccumulation (जैव संचय) / biomagnification (जैव आवर्धन) CA usually rides on pesticide or heavy-metal news; keep the mechanism pair clear.

## 3.1 Food Chain

**Identity:** Linear feeding sequence | Grazing vs detritus | 3–5 levels max

- A **food chain** is a linear sequence in which one organism is eaten by the next. Energy and nutrients pass step by step through **trophic levels**.
- The **Sun** is the ultimate energy source for almost all ecosystems. The **immediate** starting point of a chain depends on chain type.
- A **grazing (pastoral (पशुपालक)) food chain** starts from **living green plants** (producers). Energy then passes to herbivore, carnivore, and top predator.
- A **detritus food chain** starts from **dead organic matter** such as leaf litter, dead plankton, or animal carcasses. Detritivores and decomposers dominate this pathway.
- Both chain types run **in parallel** in the same ecosystem. A tiger may hunt living deer (grazing chain) and also feed on carrion (detritus chain).
- Food chains are usually **short (3–5 levels)** because only about **10%** of energy passes to the next level. Too little energy remains to support a sixth or seventh level.
- Removing one link can **break the chain** and trigger a **trophic cascade** in the levels above.
- A **terrestrial grazing chain** in UP runs Terai grass to barasingha or chital to tiger in Dudhwa.
- A classic textbook terrestrial chain is **Grass → Grasshopper → Frog → Snake** (producer → primary → secondary → tertiary consumer).
- A human-linked chain is **Grass → Goat → Human** (autotroph → herbivore → omnivore).
- An **aquatic grazing chain** runs phytoplankton to zooplankton to small fish to large fish.
- A common marine chain is **Diatoms → Crustaceans → Herrings**.

- A **detritus chain** on the Terai forest floor runs fallen sal leaves to earthworm to bacteria and fungi.
- Food chains illustrate the **order** in which organisms feed upon each other. They are **not** found within the population (जनसंख्या) of a single species.
- Humans act as **primary as well as secondary consumers** because they eat both plant products and meat.
- In the marine environment (पर्यावरण), the main primary producers are **phytoplankton** (micro-algae), not seaweeds alone.
- Main primary consumers are herbivores such as **deer**. **Ants** may also feed as primary consumers. Fox and tiger are not primary consumers.
- A large fall in butterfly populations can harm **pollination** of some plants and reduce prey for wasps, spiders, and birds. It does not by itself force a drastic rise in fungal infections of cultivated plants.

| Type | Starts from | UP / textbook example |
|---|---|---|
| **Grazing** | Living producers | Grass → deer → tiger (Dudhwa); Grass → grasshopper → frog → snake |
| **Detritus** | Dead organic matter | Leaf litter → earthworm → fungi |
| **Marine** | Phytoplankton/diatoms | Diatoms → crustaceans → herrings |
| **Parasitic** (special) | Living host tissue | Tree → insects → parasites |

> **Logic:** Grazing chain starts from a **living producer**. Detritus chain starts from **dead matter**. Trap: "all chains begin with green plants" is **false**.

## 3.2 Food Web

**Identity:** Network of chains | Stability through alternatives | Omnivores as connectors

- A **food web** is a network formed when **multiple food chains interconnect** in one ecosystem.
- Food webs represent nature more accurately than a single chain because most animals eat more than one prey species.
- Webs are **more stable** than isolated chains. If one prey species declines, predators can switch to alternatives.
- **Omnivores** such as bears, crows, and humans link grazing and detritus pathways. They can occupy **more than one trophic level** at once.
- **Keystone predators** such as tigers regulate entire webs. Removing them causes trophic cascade across several chains.
- Simpler webs in agricultural monoculture are **less stable**. Removing one pest predator can cause a pest outbreak.
- Energy still flows **unidirectionally** through every pathway in the web. A web shows many one-way paths, not cyclic energy.

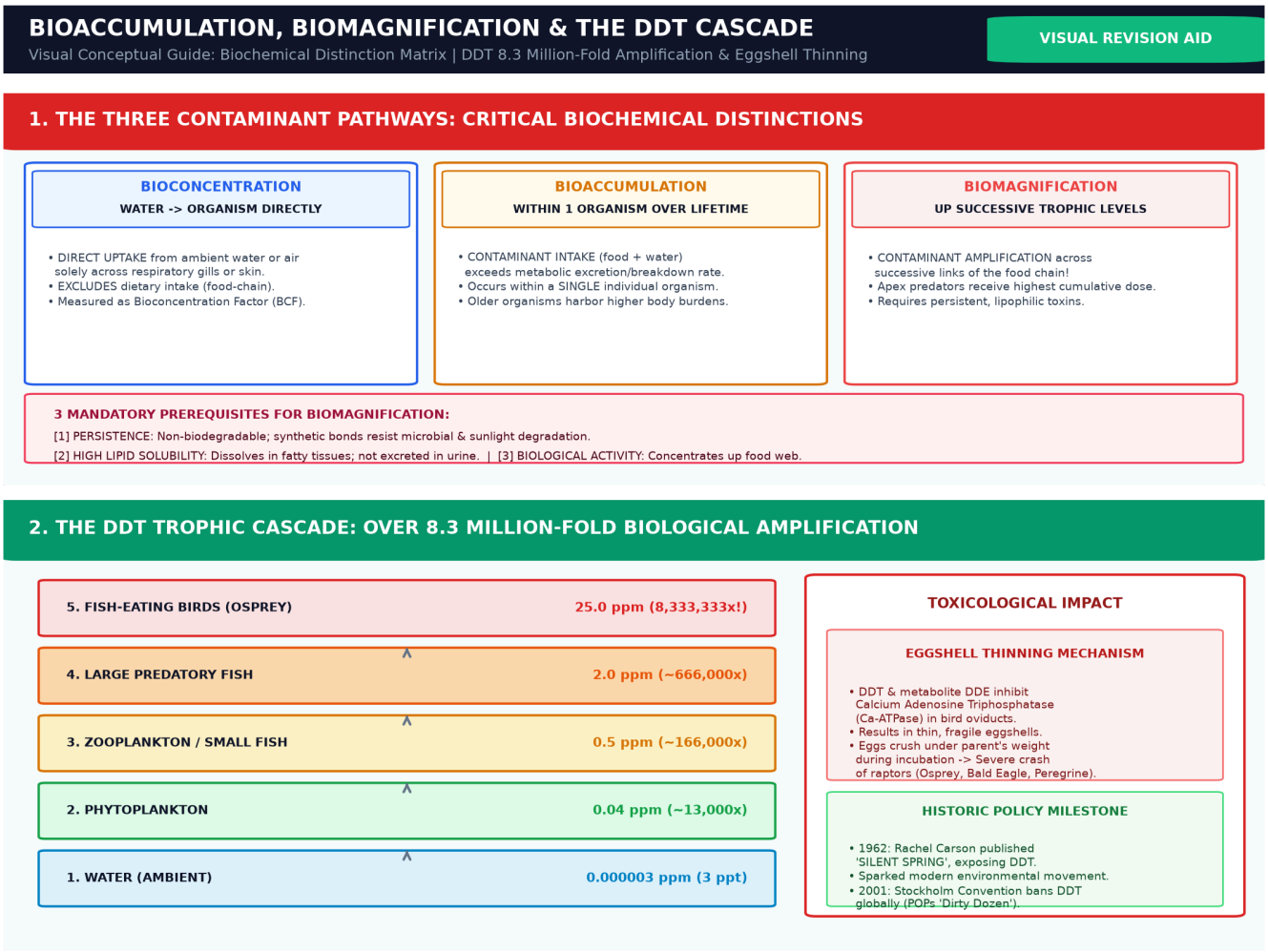
> **Logic:** Trap — "a food web makes energy flow cyclic" is **false**. Only **nutrients** cycle through decomposers.

---

## 3.3 Trophic Levels

**Identity:** T1 = producers | T2–T5 consumers | Decomposers beside the ladder

- A **trophic level** is the feeding rank of an organism in a food chain. Count from **producer = T1**. The Sun is an energy **source**, not a numbered trophic level.
- **T1 (Producers)** are autotrophs such as green plants, algae, and phytoplankton. They fix solar energy into chemical energy.
- **T2 (Primary consumers)** are herbivores that eat producers directly. Examples include deer, zooplankton, and caterpillars.
- **T3 (Secondary consumers)** are carnivores that eat primary consumers. Examples include frogs and small fish.
- **T4 (Tertiary consumers)** eat secondary consumers. Examples include snakes and large fish.
- **T5 (Quaternary consumers)** are top predators with few or no natural enemies. Tigers and eagles appear here when the chain is long enough.
- **Omnivores** eat both plants and animals. They span **multiple trophic levels** simultaneously.
- **Decomposers** break down dead matter from **all levels**. They recycle nutrients but are usually drawn **beside** the main chain, not as the top of an energy pyramid.
- Decomposer organisms in standard stems are **fungi and bacteria**. **Viruses** are not counted as decomposers here.
- **Detritivores** such as earthworms fragment dead material and bridge the detritus chain toward decomposers.




## Bioaccumulation and Biomagnification

- **Biomagnification** means a non-degradable toxin (classic example **DDT**, also mercury) rises in concentration at successive trophic levels and lodges in fatty tissues. The highest concentration appears in top consumers such as a **snake** in a grasshopper–toad–snake chain.

| Level | Role | UP / India example |
|---|---|---|
| **T1** | Producer | Terai grass, wheat, Ganga algae |
| **T2** | Primary consumer (प्राथमिक उपभोक्ता) | Barasingha, chital, rohu (juvenile) |
| **T3** | Secondary consumer (द्वितीयक उपभोक्ता) | Frog, catfish |
| **T4** | Tertiary consumer | Python, fishing eagle |
| **T5** | Top carnivore | Tiger (Dudhwa), Gangetic dolphin (river apex) |

**PYQ — UPPCS Prelims 2023, Q117**

Trophic levels are formed by—

A. Organisms linked in a food chain

B. Plants only

C. Animals only

D. Carnivorous animals only

**Show answer**

**Ans: A**

## 3.4 Energy Flow in Ecosystem

**Identity:** Unidirectional flow | Lindeman 10% | Nutrients cycle, energy does not

- **Energy flow** is the movement of solar energy through living systems via food chains and food webs.
- Flow is **unidirectional**: Sun → producers (chemical energy in biomass) → consumers → **heat** lost to the environment.
- **Nutrients** such as carbon and nitrogen **cycle** through decomposers and return to producers. **Energy does not cycle.**
- **Raymond Lindeman (1942)** showed that roughly **10%** of energy or organic matter passes to the next higher trophic level.
- About **90%** is lost at each transfer through respiration, movement, egestion, uneaten body parts, and heat radiation.
- This loss explains why **top predators are few**, why **food chains are short**, and why eating lower on the chain is more energy-efficient for humans.
- Terrestrial plants capture about **1%** of incident sunlight on leaves. That capture stage is separate from the **10% transfer** between consumer levels. Both facts can be true in one question.
- **Ecological efficiency** equals energy at the higher level divided by energy at the lower level, multiplied by 100. The NCERT standard is about **10%**.

| Where energy goes at each level | Key fact |
|---|---|
| Respiration and heat | Largest share — why pyramid of energy (ऊर्जा पिरामिड) is upright |
| Movement and metabolism | Not passed to next consumer |
| Egestion and uneaten parts | Lost from food chain pathway |
| Passed to next trophic level | **~10%** (Lindeman 1942) |

**PYQ — UPPCS Prelims 2019, Q134**

The amount of energy during transfer from one trophic level to another in an ecosystem:

A. Increases

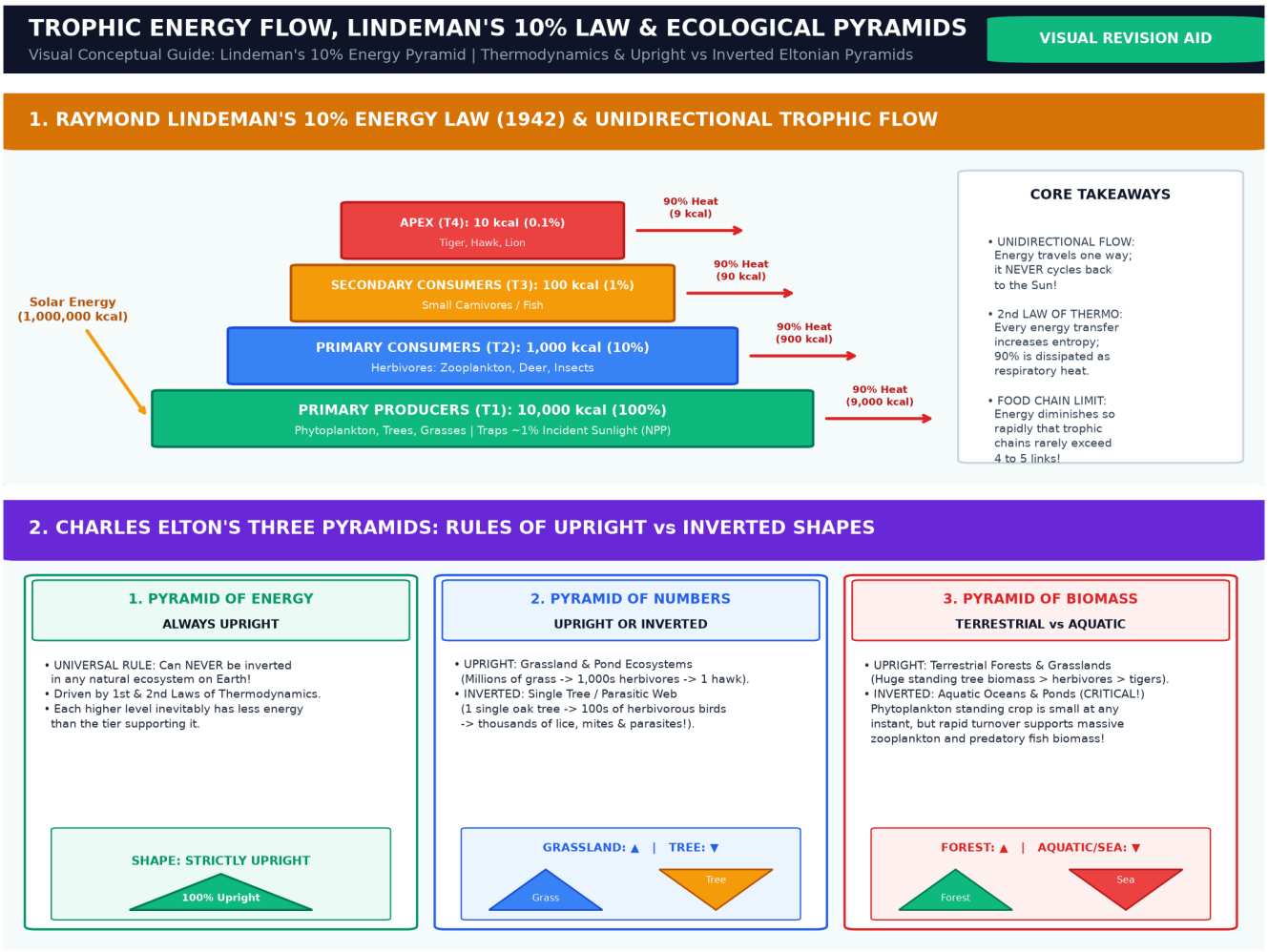
B. Decreases

C. Remains constant

D. May increase or decrease

**Show answer**

**Ans: B** — Lindeman 10% rule. Only about 10% moves up; the rest is lost.




## 3.5 Ecological Pyramid (Ecological Pyramids)

**Identity:** Elton 1927 | Energy / biomass / numbers | Only energy pyramid always upright

- An **ecological pyramid** is a diagram of trophic structure. It shows energy, biomass, or numbers at each level.
- **Charles Elton (1927)** developed the early pyramid concept in trophic studies.
- A **pyramid of energy** plots energy flow in **kcal/m²/yr**. It is **always upright** because energy is lost at every transfer.
- A **pyramid of biomass** plots standing dry weight in **g/m²**. It is usually upright on land but can be **inverted in open ocean and pond ecosystems**.
- In a **pond**, producers are tiny organisms with low standing biomass, so biomass increases toward higher consumers and the pyramid looks inverted.
- In the sea, phytoplankton reproduce so fast that at any instant their **standing biomass** may be less than fish biomass, even though energy flow still moves upward correctly.

- A **pyramid of numbers** (संख्या पिरामिड) counts **individual organisms** at each level. It is usually upright but can be **inverted in parasitic chains**.
- One large tree (one individual at T1) can support millions of insects and parasites at higher levels. That produces an inverted number pyramid.
- The **pyramid of energy** is the best measure of ecosystem function because it shows actual energy availability and is never inverted.
- **Decomposers** are usually omitted from standard pyramid diagrams. They act on all levels and return nutrients to producers.
- **Biomagnification** of toxins such as DDT increases concentration up the food chain. Top predators suffer most even as energy decreases upward.

| Pyramid | Unit | Always upright? | Inverted example |
|---|---|---|---|
| **Energy** | kcal/m²/yr | **Yes — always** | Never inverted |
| **Biomass** | g/m² | Usually | Open ocean (phytoplankton turnover) |
| **Numbers** | Individual count | Usually | One tree → many insects/parasites |

> **Logic:** Trap — "all ecological pyramids are always upright" is **false**. Only the **energy** pyramid is always upright.

---